Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Dinges, David F. Ph.D.		
Project Title:	Optical Computer Recognition of Performance Under Stress		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	02/28/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	33
No. of Bachelor's Candidates:	33	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Metaxas, Dimitris (Rutgers University)		
Grant/Contract No.:	NCC 9-58-NBPF00402		
Performance Goal No.:			
Performance Goal Text:			

This project is focused on the early objective detection of neurobehavioral problems in space flight. The overarching goal is to develop an unobtrusive, automated optical technology to detect psychological distress when astronauts are working during spaceflight. The primary aim is to determine whether optical computer recognition algorithms based on changes in facial expressions can discriminate behavioral stress induced by low versus high workload performance demands. Astronauts are required to perform mission-critical tasks at a high level of functional capability throughout spaceflight. There are a number of stressors that can compromise their ability to do so. In order to provide timely countermeasures for stressor-induced impairments in astronauts, objective, unobtrusive measures of the presence of stress reactions—especially during performance demands—are needed. This project seeks to achieve such a measurement methodology, through continued collaboration between two laboratories: one with expertise in the evaluation of

Task Description:	behavioral and physiological responses to stressful performance conditions (D. Dinges, Unit for Experimental Psychiatry, University of Pennsylvania School of Medicine), and the other with expertise in optical computer recognition of human expressions and gestures (D. Metaxas, Computational Biomedicine Imaging and Modeling Center, Rutgers University). The goal of this collaboration is to further develop and test an optically-based computer recognition algorithm of the face to reliably detect the presence of stress. The proposed computer-based recognition system builds on the research of Prof. Metaxas by utilizing automatic optical tracking of human subjects' subtle anatomical and motoric changes in facial expressions. Video input to the system is provided from experiments performed in the laboratory of Prof. Dinges. In this project we are experimentally testing the extent to which optical computer recognition algorithms based on facial expressions can detect behavioral stress during cognitive performance, both with and without facial edema. Because the facial edema that occurs in microgravity due to fluid shifts may degrade optical computer recognition of low versus high stress induced by performance demands, we are testing the following hypothesis twice in the same subjects, under both normal (no facial edema) conditions and under facial edema conditions. Hypothesis: Optical computer recognition algorithms of the face can reliably discriminate when subjects are undergoing behavioral stress associated with low versus high cognitive workload demands. In addition to testing this hypothesis, we are evaluating the extent to which the hypothesized outcome is influenced by the following potentially critical modifiers: Age; gender; ethnic background; degree of alexithymia (i.e., difficulty in recognizing and describing one's emotions); degree of subjective distress and physiological response (i.e., heart rate) during high workload performance demands; and the level of performance during difficult cognitive tasks. It is noteworthy that the test of this hypothesis allows optical algorithms to be idiosyncratic to individual subjects (i.e., no requirement of algorithm generalizability across subjects). We are also determining if optical computer recognition algorithms of facial expressions can be automated for online, real-time use in facial detection of behavioral stress during performance demands. We are also establishing the effects of low versus high behavioral stress induced by differential workload demands on performance on the WinSCAT, a cognitive test battery used by NASA JSC during space missions. This specific aim is exploratory and hypothesis-generating. A single-blind, repeated-measures cross-over trial is being used to achieve the specific aims and provide data required to test the hypotheses. A total of n=80 adults will be studied in a day-long laboratory session. The procedures for this experiment were completed in the past and preparations are underway for data acquisition in the coming year. During the last year, substantial progress has been made on increasing the sophistication of the optical computer recognition system to detect facial changes during performance stress. Significant advances in programming and enhancement of the capabilities of the computer recognition algorithm were completed, to develop a technique for robust 3-dimensional (3D) tracking of facial expression. Using this approach, numerous low-level, well-known computer vision algorithms were initially used to extract 2-dimensional (2D) information (tracking single points, edges, shading and optical flow), from video-recording of each subject's face during low and high workload performance demands. Put simply, the new technique, developed by Dr. Metaxas and colleagues, allowed the translation of 2D video footage to a form that could be used for tracking the 3D orientation and translation of the face, as well as parameters that describe the movement of eyebrows, mouth, etc. This novel technology provided us with a sensitive and unobtrusive mechanism for the measurement of changes in facial expression. Data analyses of the accuracy of the algorithm to detect performance stress are continuing on the first n=60 subjects studied. We are continuing with the development of the deformable face masks to increase the quality and precision of the tracking. The findings to date continue to support the hypothesis that an Optical Computer Recognition system can be developed to detect behavioral stress from facial expressions. In the coming year, data acquisition will begin on experiment 2. Work will also continue on improving the Optical Computer Recognition algorithm by adding facial touching gestures. Finally, work will continue on automated application of the deformable mask from video of the moving face to increase practical applicability.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study focuses on the ability of an unobtrusive, automated optical technology to detect psychological distress (and the need for countermeasures for it) during operational performance. The knowledge gained has the potential to identify an objective, unobtrusive, automated method for the recognition, monitoring, and management of the risks of neurobehavioral dysfunction due to work-related stress in spaceflight and in many Earth-based safety-sensitive occupations, such as transportation workers (e.g., truck drivers, train conductors, airline pilots); operators in safety-sensitive industries (e.g., power plant control rooms); and military personnel.
Task Progress:	With the data from experiment 1, progress was made in refining the deformable facial masks and the optical computer algorithm, to increase the sophistication of the optical computer recognition system to detect facial changes during performance stress. The final phase of the project, involving the application of the deformable masks and optical computer algorithm to video images of the subjects while performing in the low versus high workload conditions, is still underway. Experiment 1 was completed on 60 healthy adults undergoing behavioral stress induced by low versus high workload performance demands. Bootstrap development of the OCR algorithm was completed using expressive elements of the face identified by a human observer (blind to high- versus low-stressor performance demands). These parameters were used in a Hidden Markov Model to identify high- and low-stressor conditions. The OCR algorithm—developed in the Metaxas laboratory —relied on changes in mouth and eyebrow regions to accurately discriminate high- from low-stressor performance bouts in 75%-88% of subjects. Additional research begun in the later half of the past year will add facial touching gestures to further improve the OCR detection algorithm, as well as automating application of the deformable masks and OCR algorithms to video footage to increase practical utility.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Dinges DF, Rider RL, Dorrian J, McGlinchey EL, Rogers NL, Cizman Z, Goldenstein SK, Vogler C, Venkataraman S, Metaxas DN. "Optical computer recognition of facial expressions associated with stress induced by performance demands." Aviat Space Environ Med. 2005 Jun;76(6 Suppl):B172-82. PMID: 15943210 , Jun-2005

Awards	Dinges DF. " David F. Dinges: Recipient of the Decade of Behavior Research Award, American Psychological Association, 2005. " Jan-2005
Awards	Dinges DF. " David F. Dinges: The 39th Harry G. Armstrong Lecture at the 75th Annual Aviation Space Medicine Association scientific meeting, Anchorage, AK, May 2005." May-2005
Books/Book Chapters	Dinges DF, Rogers NL "The Future of Human Intelligence: Enhancing Cognitive Capability in a 24/7 World. " in "Extending intelligence : enhancement and new constructs." Ed. Patrick C. Kyllonen, Richard D. Roberts, Lazar Stankov. Mahwah, N.J. : Lawrence Erlbaum Publishers, 2006., Jan-2006
Presentation	Dinges, D. F. "Optical Computer Recognition of Performance Under Stress " Oct-2004
Presentation	Dinges, D. F., Mc Glinchey, E. L., Venkataraman, S., Metaxas, D. N. "Optical computer recognition of behavioral stress " Feb-2005
Presentation	Dinges, D. F., Mc Glinchey, E. L., Venkataraman, S., Metaxas, D. N. "Optical computer recognition of behavioral stress in space flight " Jan-2005
Presentation	Mc Glinchey, E. L., Rogers, N. L., Rider, R. L., Dorrian, J., Dinges, D. F. "Investigating the relationship between subjective assessment of sleep and mood " Jun-2004
Presentation	Mc Glinchey, E. L., Rogers, N. L., Rider, R. L., Dorrian, J., Dinges, D. F. "Investigating the relationship between subjective assessment of sleep and mood " Jul-2004
Presentation	Metaxas, D. N., Venkataraman, S., Vogler, C. "Image-Based Stress Recognition Using a Model-Based Dynamic Face Tracking System " Jul-2004